THE LEADING TERM OF THE EULER CHARACTERISTIC OF A COHERENT MODULE TWISTED BY INVERTIBLE SHEAVES IS GIVEN BY THE LENGTHS AT THE GENERIC POINTS OF THE TOP-DIMENSIONAL COMPONENTS

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Allowed standard prerequisites (editorial).

- Filtration of a coherent module with subquotients pushed forward from integral subschemes. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-coherent-filter (tag 01YF). Supplies the filtration that reduces the lemma to pushforwards of ideal sheaves.
- Additivity of the Euler characteristic and of lengths. Stacks Project, Varieties, Lemma lemma-euler-characteristic-additive (tag 08AA) and Algebra, Lemma algebra-lemma-length-additive (tag 00IV). Gives additivity of the functional E along short exact sequences.
- Numerical polynomial of the Euler characteristic and behaviour under pushforward. Stacks Project, Varieties, Lemma lemma-numerical-polynomial-from-euler (tag 0BEM); Cohomology of Schemes, Lemma coherent-lemma-relative-affine-cohomology (tag 089W). Controls the degree of the Euler characteristic polynomial and identifies the cohomology of a closed-immersion pushforward.

Source reference. Original reference 00L5: sources/stacks-chow/tags.txt, source lines 775–775. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08AA: sources/stacks-chow/tags.txt, source lines 10525–10525. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Let $\mathcal{L}_1, \ldots, \mathcal{L}_r$ be invertible $\mathcal{O}_X$-modules. Let $d = \dim(Supp(\mathcal{F}))$. Let $Z_i \subset X$ be the irreducible components of $Supp(\mathcal{F})$ of dimension d. Let $\xi_i \in Z_i$ be the generic point and set $m_i = length_{\mathcal{O}_{X,\xi_i}}(\mathcal{F}_{\xi_i})$. Then*

$$\chi(X, \mathcal{F} \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}) - \sum\nolimits_i m_i \ \chi(Z_i, \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}|_{Z_i})$$

is a numerical polynomial in $n_1, \ldots, n_r$ of total degree $< d$.

Proof. Consider pairs (ξ, Z) where $Z \subset X$ is an integral closed subscheme of dimension d and ξ is its generic point. Then the finite $\mathcal{O}_{X,\xi}$-module $\mathcal{F}_\xi$ has support contained in $\{\xi\}$ hence the length $m_Z = \text{length}_{\mathcal{O}_{X,\xi}}(\mathcal{F}_\xi)$ is finite (Algebra, Lemma 00L5) and zero unless $Z = Z_i$ for some i. Thus the expression of the lemma can be written as

$$E(\mathcal{F}) = \chi(X, \mathcal{F} \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}) - \sum m_Z \ \chi(Z, \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}|_Z)$$

where the sum is over integral closed subschemes $Z \subset X$ of dimension d. The assignment $\mathcal{F} \mapsto E(\mathcal{F})$ is additive in short exact sequences $0 \to \mathcal{F} \to \mathcal{F}' \to \mathcal{F}'' \to 0$ of coherent $\mathcal{O}_X$-modules whose support has dimension $\leq d$. This follows from additivity of Euler characteristics (Lemma 08AA) and additivity of lengths (Algebra, Lemma 00IV). Let us apply Cohomology of Schemes, Lemma 01YF to find a filtration

$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_m = \mathcal{F}$$

by coherent subsheaves such that for each $j = 1, \ldots, m$ there exists an integral closed subscheme $V_j \subset X$ and a nonzero sheaf of ideals $\mathcal{I}_j \subset \mathcal{O}_{V_j}$ such that

$$\mathcal{F}_j/\mathcal{F}_{j-1} \cong (V_j \to X)_*\mathcal{I}_j$$

It follows that $V_j \subset \text{Supp}(\mathcal{F})$ and hence $\dim(V_j) \leq d$. By the additivity we remarked upon above it suffices to prove the result for each of the subquotients $\mathcal{F}_j/\mathcal{F}_{j-1}$. Thus it suffices to prove the result when $\mathcal{F} = (V \to X)_*\mathcal{I}$ where $V \subset X$ is an integral closed subscheme of dimension $\leq d$ and $\mathcal{I} \subset \mathcal{O}_V$ is a nonzero coherent sheaf of ideals. If $\dim(V) < d$ and more generally for $\mathcal{F}$ whose support has dimension $< d$, then the first term in $E(\mathcal{F})$ has total degree $< d$ by Lemma 0BEM and the second term is zero. If $\dim(V) = d$, then we can use the short exact sequence

$$0 \to (V \to X)_*\mathcal{I} \to (V \to X)_*\mathcal{O}_V \to (V \to X)_*(\mathcal{O}_V/\mathcal{I}) \to 0$$

The result holds for the middle sheaf because the only Z occurring in the sum is $Z = V$ with $m_Z = 1$ and because

$$H^i(X, ((V \to X)_*\mathcal{O}_V) \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}) = H^i(V, \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}|_V)$$

by the projection formula (Cohomology, Section 01E6) and Cohomology of Schemes, Lemma 089W; so in this case we actually have $E(\mathcal{F}) = 0$. The result holds for the sheaf on the right because its support has dimension $< d$. Thus the result holds for the sheaf on the left and the lemma is proved. □